# Sankalp Jha

Backend Engineer - Go, Distributed Systems & Cloud Infrastructure
Delhi NCR, India

**Portfolio** | **LinkedIn** | **GitHub**
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## Education

| Year | Degree/Certificate | Institute | CPI/% |
|---|---|---|---|
| **Oct 2022 – Aug 2026** | B.Tech in CS & IT | Ajay Kumar Garg Engineering College, Delhi NCR | **7.25/10** |
| **2021 – 2022** | Senior Secondary (XII) | New Era School, Delhi NCR | **91.2%** |

## Experience

**Google Summer of Code 2025 Contributor** | *P4 Language Consortium, Stanford, CA* (***May 2025 – Sept 2025***)

**SpliDT Project: Scaling Stateful Decision Tree Algorithms in P4** *P4, Python, P4Runtime, Docker, C++, Intel Tofino*

- **Switch-Native Architecture:** Compiled ML decision trees directly into P4 data planes with TCAM-efficient Subtree IDs to resolve hardware ASIC constraints.
- **Validation & Optimization:** Built a P4Runtime testing pipeline (Tofino/Mininet) improving reliability by **30%**; hit **10** Gbps line rate with **20%** lower inference latency.

## Open Source Contributions

- **Project-HAMi/HAMi:** **#2292** Corrected hardware allocation logic for Kunlun and AWS Neuron AI accelerators to prevent invalid state generation; **#2307** Eliminated node-level deadlocks by strictly enforcing lock releases during failed device-binding operations; **#2204** Prevented metrics panics by guarding hardware memory ratio calculations against unknown total memory states; **#2259** Stabilized cluster scheduling by adding safe state handling for deleted ResourceQuota tombstones.
- **WasmEdge/WasmEdge:** **#4470** fix(wasi): resolve **100%** CPU usage on macOS due to clock mismatch; **#4466** feat(docker): add standalone alpine-base image build pipeline.
- **facebook/bpfilter:** **#507** build: update clang-tidy brace config, fix opts.c/set.c, and NOLINT dump.c.

## Key Projects

**Myprod** | *Go, Nomad, Traefik, WireGuard, Docker, Netlify DNS, SOPS* (*Project Link*)

**Objective:** Turn manually provisioned VPS machines into a low-overhead personal compute pool for backend services.
**Approach:** Built a Go CLI and operator dashboard around Nomad scheduling, Traefik ingress, and WireGuard networking; added guarded node lifecycle controls, application registration, live resource metrics, and idempotent Netlify DNS management.
**Impact:** Provides one control surface for deploying applications and managing heterogeneous free-tier and temporary compute capacity.

**Cutable** | *Go, Next.js, PostgreSQL, OpenRouter, E2B* (*Project Link*)

**Objective:** Generate editable React applications from text and visual requirements.
**Approach:** Orchestrated OpenRouter tools and E2B sandboxes in Go; streamed previews over WebSockets.
**Impact:** Ships multimodal BYOK generation and health-gated deployments.

**DMIE** | *Next.js, Python, FastAPI, Vector DB, LLM* (*Project Link*)

**Objective:** Determine whether a startup idea already exists by comparing it against more than **500,000** company records.
**Approach:** Combined LLM query with vector search and backend filters over a scheduled scraper.
**Impact:** Raised USD **10,000** and serves more than **100** users worldwide.

**Do-It** | *Go, SSE, HTTP, Raspberry Pi, Local-first* (*Project Link*)

**Objective:** Provide private, synchronized notes and media across devices on a local network.
**Approach:** Packaged a lightweight Go SSE and HTTP server for Raspberry Pi, routers, and repurposed phones.
**Impact:** Delivers self-hosted, account-free synchronization without sending household data to a cloud service.

## Technical Skills

- **Languages:** Go, Python, C, C++, TypeScript, JavaScript
- **Distributed Systems:** Nomad, Kubernetes, Traefik, WireGuard, gRPC, Protobufs, Docker
- **Data & Storage:** PostgreSQL, Redis, MongoDB, MySQL, SQL, GraphQL
- **Tools / Platforms:** CI/CD, GitHub Actions, Linux (x86_64, ARM64), SOPS, Netlify DNS

## Positions of Leadership

- **Technical Writer, Cloud Native Community Group, New Delhi** (***May 2024 – Dec 2024***)
  - Led **6+** writers and **4** designers on documentation and event content; social media revamp lifted engagement **60%**.
- **Co-Organizer & Anchor, Google Developer Groups & GDSC WOW 2024, Delhi NCR** (***Nov 2023 – Nov 2025***)
  - Co-led **10+** tech events; anchored GDSC WOW at IIIT Delhi (**800+** live audience, **31+** crew).

## Publications

- R. Bharadwaj, **S. Jha**, V. Malyan, R. Gupta, "Trusture: A Blockchain-Based Decentralized Framework for Secure, Transparent, and Auditable NGO Transactions," ***2026*** *ICCSC*, Ghaziabad, India, Feb. **2026**. DOI: 10.1109/ICCSC67078.2026.11468597

## Certifications & Honors

- **2x** AKTU State TT Gold (**2023**, **2024**); Robotics Olympiad Bronze, Thailand (**2019**); NASA, NPTEL & ISRO certifications